



Project 1 Hardening Summary and Checklist

OS Information

Customer	Baker Street Corporation														
Hostname	<u>Baker_Street_Linux_Server</u>														
OS Version	<u>"Ubuntu 22.04.5 LTS"</u>														
Memory information	<table><tr><td></td><td>total</td><td>used</td><td>free</td><td>shared</td><td>buff/cache</td><td>available</td></tr><tr><td>Mem:</td><td>15Gi</td><td>1.3Gi</td><td>10Gi</td><td>201Mi</td><td>3.2Gi</td><td>13Gi</td></tr></table>		total	used	free	shared	buff/cache	available	Mem:	15Gi	1.3Gi	10Gi	201Mi	3.2Gi	13Gi
	total	used	free	shared	buff/cache	available									
Mem:	15Gi	1.3Gi	10Gi	201Mi	3.2Gi	13Gi									
Uptime information	<u>00:30:16 up 57 min, 0 users, load average: 0.14, 0.20, 0.23</u>														

Checklist

Completed	Activity	Script(s) used / Tasks completed / Screenshots



OS backup

```
root@Baker_Street_Linux_Server: /
File Edit View Search Terminal Help
sysadmin@ip-10-0-1-138:~$ docker exec -it project1_v4 /bin/bash
root@Baker_Street_Linux_Server:/# sudo tar -cvpzf /baker_street_backup.tar.gz --
exclude=/baker_street_backup.tar.gz --exclude=/proc --exclude=/tmp --exclude=/mn
t --exclude=/sys --exclude=/dev --exclude=/run /

sudo: unable to resolve host Baker_Street_Linux_Server: Temporary failure in nam
e resolution
tar: Removing leading '/' from member names
/
/bin
/home/
/home/lestrade/
/home/lestrade/.bashrc
/home/lestrade/.bash_logout
/home/lestrade/.profile
/home/lestrade/Engineering_script.sh_script1.sh
/home/lestrade/Research_script.sh_script1.sh
/home/lestrade/Engineering_script.sh_script2.sh
/home/lestrade/Research_script.sh_script2.sh
/home/lestrade/Research_script.sh_2.txt
/home/lestrade/Engineering_script.sh_0.txt
/home/lestrade/Research_script.sh_3.txt
/home/lestrade/deduction.doc_1.txt
/home/adler/
```

```
root@Baker_Street_Linux_Server: /
File Edit View Search Terminal Help
/etc/rpc
/etc/ufw/
/etc/ufw/applications.d/
/etc/ufw/applications.d/samba
/etc/ufw/applications.d/openssh-server
/etc/ca-certificates.conf
/etc/perl/
/etc/perl/Net/
/etc/perl/Net/libnet.cfg
/etc/ethertypes
/etc/cron.hourly/
/etc/cron.hourly/.placeholder
/etc/dbus-1/
/etc/dbus-1/system.d/
/etc/dbus-1/session.d/
/etc/python3.10/
/etc/python3.10/sitecustomize.py
/boot/
/media/
/lib32
/sbin
/.dockerenv
tar: /: file changed as we read it
root@Baker_Street_Linux_Server:/#
```



Auditing users and groups

```
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
_apt:x:100:65534::/nonexistent:/usr/sbin/nologin
systemd-network:x:101:102:systemd Network Management,,:/run/systemd:/usr/
systemd-resolve:x:102:103:systemd Resolver,,:/run/systemd:/usr/sbin/nolog
mysql:x:103:104:MySQL Server,,:/nonexistent:/bin/false
messagebus:x:104:106::/nonexistent:/usr/sbin/nologin
systemd-timesync:x:105:107:systemd Time Synchronization,,:/run/systemd:/u
syslog:x:106:108::/home/syslog:/usr/sbin/nologin
sshd:x:107:65534::/run/ssh:/usr/sbin/nologin
sherlock:x:1000:1000::/home/sherlock:/bin/bash
watson:x:1001:1001::/home/watson:/bin/bash
moriarty:x:1002:1002::/home/moriarty:/bin/bash
mycroft:x:1003:1003::/home/mycroft:/bin/bash
mrs_hudson:x:1006:1006::/home/mrs_hudson:/bin/bash
sysadmin:x:1008:1008::/home/sysadmin:/bin/bash
toby:x:1010:1010::/home/toby:/bin/bash
adler:x:1011:1011::/home/adler:/bin/bash
root@Baker_Street_Linux_Server:/etc#
```

```
root@Baker_Street_Linux_Server:/etc# cd ..
root@Baker_Street_Linux_Server:/# getent shadow | grep '!t'
root@Baker_Street_Linux_Server:/# getent shadow | grep '!'
mysql:!:20069:0:99999:7:::
moriarty:!:sysj9T$2ar6xmt30H5w0H8xnuDa/$xtdB/1L.LQDvVP50kjGHeLo5LVUAp9De3Z106PXH
N53:20074:0:99999:7:::
mrs_hudson:!:20069:0:99999:7:::
mary:!:20069:0:99999:7:::
sysadmin:!:20069:0:99999:7:::
gregson:!:20069:0:99999:7:::
root@Baker_Street_Linux_Server:/#
```

Locked and removed users

```
root@Baker_Street_Linux_Server:/etc
File Edit View Search Terminal Help
mrs_hudson:x:1006:1006::/home/mrs_hudson:/bin/bash
sysadmin:x:1008:1008::/home/sysadmin:/bin/bash
toby:x:1010:1010::/home/toby:/bin/bash
adler:x:1011:1011::/home/adler:/bin/bash
root@Baker_Street_Linux_Server:/etc# cd ..
root@Baker_Street_Linux_Server:/# ls
bin  dev  home  lib32  libx32  mnt  proc  run  srv  tmp  var
boot  etc  lib  lib64  media  mps  root  sbin  sys  usr
root@Baker_Street_Linux_Server:/# cd /home
root@Baker_Street_Linux_Server:/home# ls
adler  irene  mary  mrs_hudson  sherlock  toby
gregson  lestrade  moriarty  mycroft  sysadmin  watson
root@Baker_Street_Linux_Server:/home# deluser mary/
/usr/sbin/deluser: The user 'mary/' does not exist.
root@Baker_Street_Linux_Server:/home# cd ..
root@Baker_Street_Linux_Server:/# userdel mary
userdel: user 'mary' does not exist
root@Baker_Street_Linux_Server:/# sudo usermod -L moriarty
sudo: unable to resolve host Baker_Street_Linux_Server: Temporary failure in name resolution
root@Baker_Street_Linux_Server:/# usermod -L moriarty
root@Baker_Street_Linux_Server:/# usermod -L mrs_hudson
root@Baker_Street_Linux_Server:/# usermod -U sherlock
root@Baker_Street_Linux_Server:/# usermod -U watson
root@Baker_Street_Linux_Server:/# usermod -U mycroft
root@Baker_Street_Linux_Server:/# usermod -U toby
usermod: unlocking the user's password would result in a passwordless account.
You should set a password with usermod -p to unlock this user's password.
root@Baker_Street_Linux_Server:/# usermod p toby
Usage: usermod [options] LOGIN
Options:
```

```
root@Baker_Street_Linux_Server:~# cd etc
root@Baker_Street_Linux_Server:~/etc# nano shadow
root@Baker_Street_Linux_Server:~/etc# passwd toby
New password:
Re-type new password:
passwd: password updated successfully
root@Baker_Street_Linux_Server:~/etc# cd ..
root@Baker_Street_Linux_Server:~# usermod -U toby
root@Baker_Street_Linux_Server:~# usermod -U adler
usermod: unlocking the user's password would result in a passwordless account.
You should set a password with usermod -p to unlock this user's password.
root@Baker_Street_Linux_Server:~# passwd adler
New password:
Re-type new password:
passwd: password updated successfully
root@Baker_Street_Linux_Server:~# usermod -U adler
root@Baker_Street_Linux_Server:~#
```

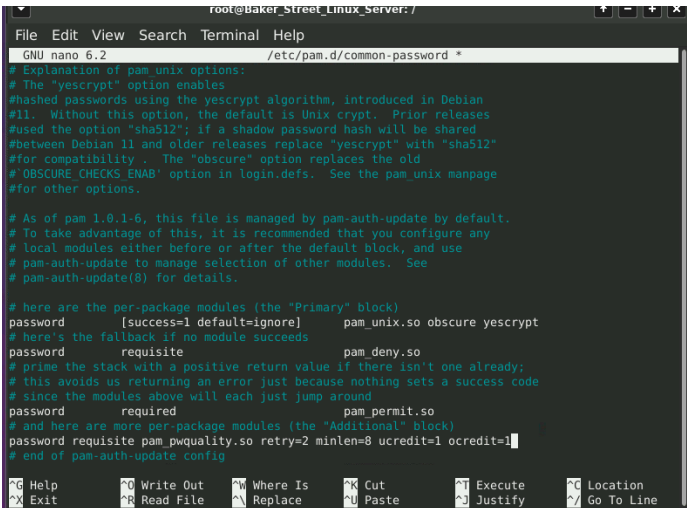
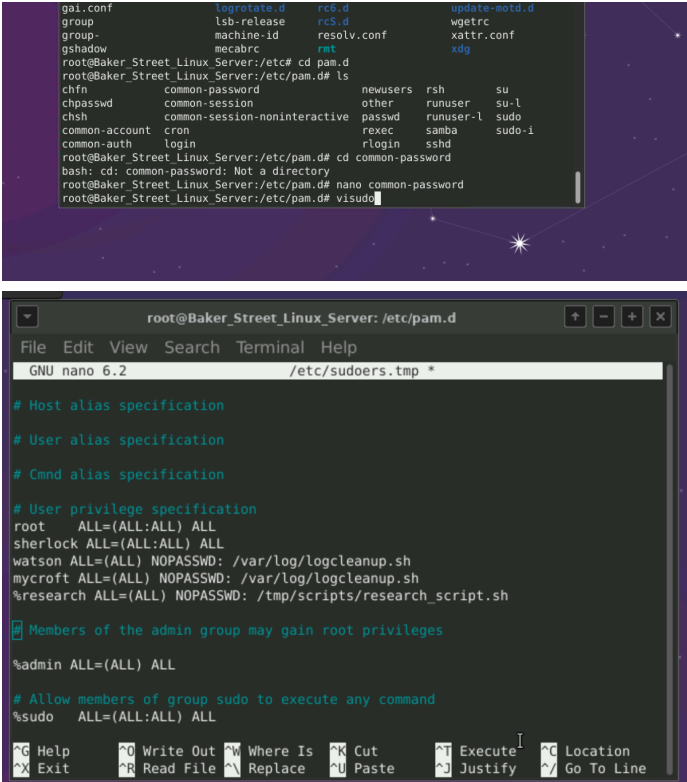
Issue with container deleting users. Had to reboot.

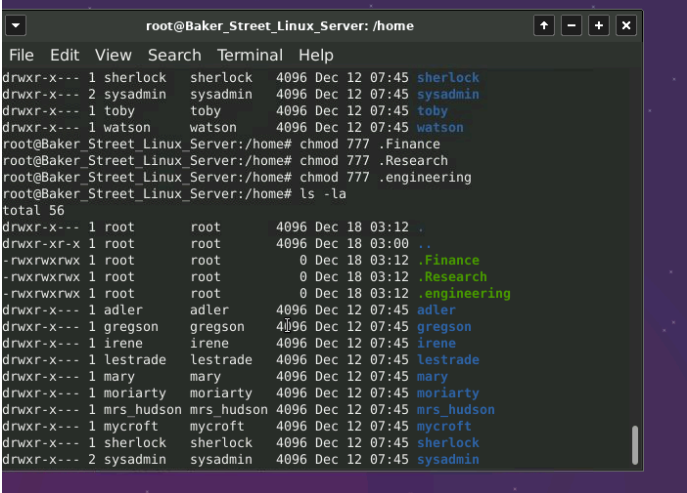
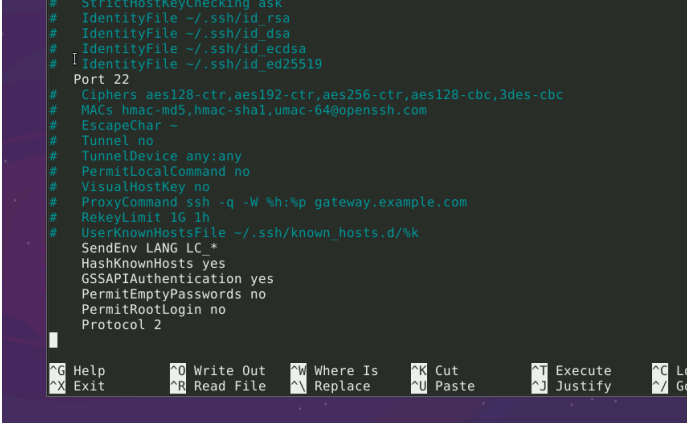
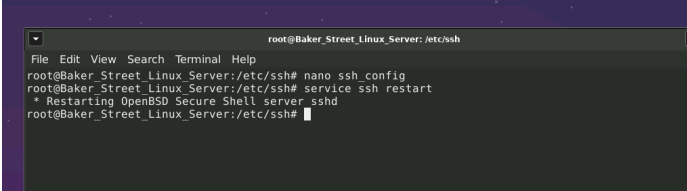
Add users to research group

```
root@Baker_Street_Linux_Server:~# groupadd research
root@Baker_Street_Linux_Server:~# usermod -aG research sherlock
root@Baker_Street_Linux_Server:~# usermod -aG research adler
root@Baker_Street_Linux_Server:~# usermod -aG research watson
root@Baker_Street_Linux_Server:~# usermod -aG research mycroft
root@Baker_Street_Linux_Server:~# usermod -aG research toby
root@Baker_Street_Linux_Server:~# cd etc
root@Baker_Street_Linux_Server:~/etc# cat group
root:x:0:
daemon:x:1:
bin:x:2:
sys:x:3:
adm:x:4:syslog
tty:x:5:
disk:x:6:
```

```
mycroft:x:1003:
irene:x:1004:
lestrade:x:1005:
mrs_hudson:x:1006:
mary:x:1007:
sysadmin:x:1008:
gregson:x:1009:
toby:x:1010:
adler:x:1011:
engineering:x:1012:sherlock,watson,moriarty
finance:x:1013:mrs_hudson
marketing:x:1014:
research:x:1015:sherlock,adler,watson,mycroft,toby
root@Baker_Street_Linux_Server:~/etc#
```

```
root@Baker_Street_Linux_Server:~# cd home
root@Baker_Street_Linux_Server:~/home# ls
adler gregson irene lestrade mary moriarty mrs_hudson mycroft sherlock sysadmin toby watson
root@Baker_Street_Linux_Server:~/home# userdel irene
root@Baker_Street_Linux_Server:~/home# userdel lestrade/
userdel: user lestrade/ does not exist
root@Baker_Street_Linux_Server:~/home# userdel lestrade
root@Baker_Street_Linux_Server:~/home# awk -F: '($7 != "/usr/sbin/nologin" && $7 != "/bin/false") {etc/passwd
| awk -F: 'NR==FNR {active[$1]; next} $1 in active && $2 != "/" {print $1}' - /etc/shadow
sherlock:sys:9T57iE18VdxLVZiu7Z6Awrk/$XVP88f404e/LD9KUHJ5uvW5KW82IRp.Rlg7hu1JF39:20076:0:99999:7:::
watson:sys:9T56/Dov83YG0DuK1J7cZV.S0KNSK9u0eWdIat8KynVQ6G7HhWjgxEUUCCompL4:20076:0:99999:7:::
moriarty:sys:9T51m7WetpYInGALxvFns1pajNkZub0Eqrl.kTfg19Kdvc0op0oeUN3pk1C7DITMC:20076:0:99999:7:::
mycroft:sys:9T58da5xerWdbLqx/Ido2/8J/sF1BAGEHKqr/L3dJ69qRKTXgnDzPPFAz9n1g1G1LRMu7:20076:0:99999:7:::
root@Baker_Street_Linux_Server:~/home#
```

☒	Updating and enforcing password policies	 <pre> root@Baker_Street_Linux_Server: / File Edit View Search Terminal Help GNU nano 6.2 /etc/pam.d/common-password * # Explanation of pam_unix options: # The "yescrypt" option enables # hashed passwords using the yescrypt algorithm, introduced in Debian # 11. Without this option, the default is Unix crypt. Prior releases # used the option "sha512"; if a shadow password hash will be shared # between Debian 11 and older releases replace "yescrypt" with "sha512" # for compatibility. The "obscure" option replaces the old # "OBSOLETE_CHECKS_ENAB" option in login.defs. See the pam_unix manpage # for other options. # As of pam 1.0.1-6, this file is managed by pam-auth-update by default. # To take advantage of this, it is recommended that you configure any # local modules either before or after the default block, and use # pam-auth-update to manage selection of other modules. See # pam-auth-update(8) for details. # here are the per-package modules (the "Primary" block) password [success=1 default=ignore] pam_unix.so obscure yescrypt # here's the fallback if no module succeeds password requisite pam_deny.so # prime the stack with a positive return value if there isn't one already; # this avoids us returning an error just because nothing sets a success code # since the modules above will each just jump around password required pam_permit.so # and here are more per-package modules (the "Additional" block) password requisite pam_pwquality.so retry=2 minlen=8 ucredit=1 ocredit=1 # end of pam-auth-update config </pre> 2 retries 8 minimum length of passwd 1 uppercase letter 1 special character
☒	Updating and enforcing sudo permissions	 <pre> gai.conf logrotate.d rc6.d update-motd.d group lsb-release rc5.d wgetrc group machine-id resolv.conf xattr.conf gshadow mecabrc rmt xdg root@Baker_Street_Linux_Server:/etc# cd pam.d root@Baker_Street_Linux_Server:/etc/pam.d# ls chfn common-password newusers rsh su chpasswd common-session other runuser su-l chsh common-session-noninteractive passwd runuser-l sudo common-account cron rexec samba sudo-i common-auth login rlogin sshd root@Baker_Street_Linux_Server:/etc/pam.d# cd common-password bash: cd: common-password: Not a directory root@Baker_Street_Linux_Server:/etc/pam.d# nano common-password root@Baker_Street_Linux_Server:/etc/pam.d# visudo </pre> <pre> root@Baker_Street_Linux_Server: /etc/pam.d File Edit View Search Terminal Help GNU nano 6.2 /etc/sudoers.tmp * # Host alias specification # User alias specification # Cmnd alias specification # User privilege specification root ALL=(ALL:ALL) ALL sherlock ALL=(ALL:ALL) ALL watson ALL=(ALL) NOPASSWD: /var/log/logcleanup.sh mycroft ALL=(ALL) NOPASSWD: /var/log/logcleanup.sh %research ALL=(ALL) NOPASSWD: /tmp/scripts/research_script.sh Members of the admin group may gain root privileges %admin ALL=(ALL) ALL # Allow members of group sudo to execute any command %sudo ALL=(ALL:ALL) ALL </pre>

☒	Validating and updating permissions on files and directories	 <pre> root@Baker_Street_Linux_Server: /home File Edit View Search Terminal Help drwxr-x--- 1 sherlock sherlock 4096 Dec 12 07:45 sherlock drwxr-x--- 2 sysadmin sysadmin 4096 Dec 12 07:45 sysadmin drwxr-x--- 1 toby toby 4096 Dec 12 07:45 toby drwxr-x--- 1 watson watson 4096 Dec 12 07:45 watson root@Baker_Street_Linux_Server:/home# chmod 777 .Finance root@Baker_Street_Linux_Server:/home# chmod 777 .Research root@Baker_Street_Linux_Server:/home# chmod 777 .engineering root@Baker_Street_Linux_Server:/home# ls -la total 56 drwxr-x--- 1 root root 4096 Dec 18 03:12 . drwxr-xr-x 1 root root 4096 Dec 18 03:00 .. -rwxrwxrwx 1 root root 0 Dec 18 03:12 .Finance -rwxrwxrwx 1 root root 0 Dec 18 03:12 .Research -rwxrwxrwx 1 root root 0 Dec 18 03:12 .engineering drwxr-x--- 1 adler adler 4096 Dec 12 07:45 adler drwxr-x--- 1 gregson gregson 4096 Dec 12 07:45 gregson drwxr-x--- 1 irene irene 4096 Dec 12 07:45 irene drwxr-x--- 1 lestrade lestrade 4096 Dec 12 07:45 lestrade drwxr-x--- 1 mary mary 4096 Dec 12 07:45 mary drwxr-x--- 1 moriarty moriarty 4096 Dec 12 07:45 moriarty drwxr-x--- 1 mrs_hudson mrs_hudson 4096 Dec 12 07:45 mrs_hudson drwxr-x--- 1 mycroft mycroft 4096 Dec 12 07:45 mycroft drwxr-x--- 1 sherlock sherlock 4096 Dec 12 07:45 sherlock drwxr-x--- 2 sysadmin sysadmin 4096 Dec 12 07:45 sysadmin </pre>
☐	Optional: Updating password hashing configuration	
☒	Auditing and securing SSH	 <pre> # StrictHostKeyChecking ask # IdentityFile ~/.ssh/id_rsa # IdentityFile ~/.ssh/id_dsa # IdentityFile ~/.ssh/id_ecdsa # IdentityFile ~/.ssh/id_ed25519 Port 22 # Ciphers aes128-ctr,aes192-ctr,aes256-ctr,aes128-cbc,3des-cbc # MACs hmac-md5,hmac-sha1,umac-64@openssh.com # EscapeChar ~ # Tunnel no # TunnelDevice any:any # PermitLocalCommand no # VisualHostKey no # ProxyCommand ssh -q -W %h:%p gateway.example.com # RekeyLimit 1G 1h # UserKnownHostsFile ~/.ssh/known_hosts.d/%k SendEnv LANG LC * HashKnownHosts yes GSSAPIAuthentication yes PermitEmptyPasswords no PermitRootLogin no Protocol 2 </pre>  <pre> root@Baker_Street_Linux_Server: /etc/ssh File Edit View Search Terminal Help root@Baker_Street_Linux_Server:/etc/ssh# nano ssh_config root@Baker_Street_Linux_Server:/etc/ssh# service ssh restart * Restarting OpenBSD Secure Shell server sshd root@Baker_Street_Linux_Server:/etc/ssh# </pre> Permitted root ability to ssh Permitted ability to ssh with empty passwords Enabled Port 22



Reviewing and updating system packages

```
root@Baker_Street_Linux_Server: /etc/ssh
File Edit View Search Terminal Help
root@Baker_Street_Linux_Server:/etc/ssh# nano ssh config
root@Baker_Street_Linux_Server:/etc/ssh# service ssh restart
* Restarting OpenBSD Secure Shell server sshd
root@Baker_Street_Linux_Server:/etc/ssh# apt update
0% [Connecting to archive.ubuntu.com]
```

```
root@Baker_Street_Linux_Server: /etc/ssh
File Edit View Search Terminal Help
root@Baker_Street_Linux_Server:/etc/ssh# nano ssh config
root@Baker_Street_Linux_Server:/etc/ssh# service ssh restart
* Restarting OpenBSD Secure Shell server sshd
root@Baker_Street_Linux_Server:/etc/ssh# apt update
Get:1 http://archive.ubuntu.com/ubuntu jammy InRelease [270 kB]
Get:2 http://archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:3 http://archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:4 http://archive.ubuntu.com/ubuntu jammy/main amd64 Packages [1792 kB]
Get:5 http://archive.ubuntu.com/ubuntu jammy/restricted amd64 Packages [164 kB]
Get:6 http://archive.ubuntu.com/ubuntu jammy/universe amd64 Packages [17.5 MB]
Get:7 http://archive.ubuntu.com/ubuntu jammy/multiverse amd64 Packages [266 kB]
Get:8 http://archive.ubuntu.com/ubuntu jammy-updates/multiverse amd64 Packages [53.3 kB]
Get:9 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [2817 kB]
Get:10 http://archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 Packages [3588 kB]
Get:11 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 Packages [1517 kB]
Get:12 http://archive.ubuntu.com/ubuntu jammy-backports/main amd64 Packages [111 kB]
Get:13 http://archive.ubuntu.com/ubuntu jammy-backports/universe amd64 Packages [37.0 kB]
Get:14 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Get:15 http://security.ubuntu.com/ubuntu jammy-security/multiverse amd64 Packages [45.2 kB]
Get:16 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 Packages [1226 kB]
Get:17 http://security.ubuntu.com/ubuntu jammy-security/restricted amd64 Packages [3436 kB]
Get:18 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [2506 kB]
Fetched 35.7 MB in 26s (1387 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
root@Baker_Street_Linux_Server:/etc/ssh#
```

```
Get:16 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 Packages [1226
Get:17 http://security.ubuntu.com/ubuntu jammy-security/restricted amd64 Packages [34
Get:18 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [2506 kB]
Fetched 35.7 MB in 26s (1387 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
root@Baker_Street_Linux_Server:/etc/ssh# apt upgrade -y
bash: apt: command not found
root@Baker_Street_Linux_Server:/etc/ssh# apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
root@Baker_Street_Linux_Server:/etc/ssh#
```

Updated packages

```
bin dev home lib32 libx32 mnt proc run srv var
boot etc lib lib64 media opt root/sbin sys usr
root@Baker_Street_Linux_Server:/# touch package_list.txt
root@Baker_Street_Linux_Server:/# ls
bin dev home lib32 libx32 mnt package_list.txt root/sbin sys usr
boot etc lib lib64 media opt proc run srv var
root@Baker_Street_Linux_Server:/# apt list --installed > package_list.txt

WARNING: apt does not have a stable CLI interface. Use with caution in scripts.

root@Baker_Street_Linux_Server:/# cat package_list.txt
Listing...
adduser/jammy,now 3.118ubuntu5 all [installed]
apt/jammy-updates,now 2.4.13 amd64 [installed]
attr/jammy,now 1:2.5.1-1build1 amd64 [installed,automatic]
base-files/jammy-updates,now 12ubuntu4.7 amd64 [installed]
base-passwd/jammy,now 3.5.52build1 amd64 [installed]
bash/jammy-updates,jammy-security,now 5.1-6ubuntu1.1 amd64 [installed]
bsdutils/jammy-updates,jammy-security,now 1:2.37.2-4ubuntu3.4 amd64 [installed]
ca-certificates/jammy-updates,jammy-security,now 20240203-22.04.1 all [installed,automatic]
```

Created file, added apt list.

```
root@Baker_Street_Linux_Server:~# apt remove rsh-client
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages will be REMOVED:
  rsh-client
0 upgraded, 0 newly installed, 1 to remove and 0 not upgraded.
After this operation, 109 kB disk space will be freed.
Do you want to continue? [Y/n] y
(Reading database ... 16312 files and directories currently installed.)
Removing rsh-client (0.17-22) ...
update-alternatives: using /usr/bin/scp to provide /usr/bin/rcp (rcp) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/rcp.1.gz because associated file /usr/sh
are/man/man1/scp.1.gz (of link group rcp) doesn't exist
update-alternatives: using /usr/bin/ssh to provide /usr/bin/rsh (rsh) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/rsh.1.gz because associated file /usr/sh
are/man/man1/ssh.1.gz (of link group rsh) doesn't exist
update-alternatives: using /usr/bin/slogin to provide /usr/bin/rlogin (rlogin) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/rlogin.1.gz because associated file /usr
/share/man/man1/slogin.1.gz (of link group rlogin) doesn't exist
root@Baker_Street_Linux_Server:~#
```

Rsh-client removed, telnet removed.

```
/share/man/man1/slogin.1.gz (of link group rlogin) doesn't exist
root@Baker_Street_Linux_Server:~# apt remove telnet
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages will be REMOVED:
  telnet
0 upgraded, 0 newly installed, 1 to remove and 0 not upgraded.
After this operation, 158 kB disk space will be freed.
Do you want to continue? [Y/n] y
(Reading database ... 16301 files and directories currently installed.)
Removing telnet (0.17-44build1) ...
root@Baker_Street_Linux_Server:~#
```

```
/share/man/man1/slogin.1.gz (of link group rlogin) doesn't exist
root@Baker_Street_Linux_Server:~# apt remove telnet
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages will be REMOVED:
  telnet
0 upgraded, 0 newly installed, 1 to remove and 0 not upgraded.
After this operation, 158 kB disk space will be freed.
Do you want to continue? [Y/n] y
(Reading database ... 16301 files and directories currently installed.)
Removing telnet (0.17-44build1) ...
root@Baker_Street_Linux_Server:~# apt autoremove -y.
E: Command line option '.' [from -y.] is not understood in combination with the other opt
root@Baker_Street_Linux_Server:~# apt autoremove -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
root@Baker_Street_Linux_Server:~#
```

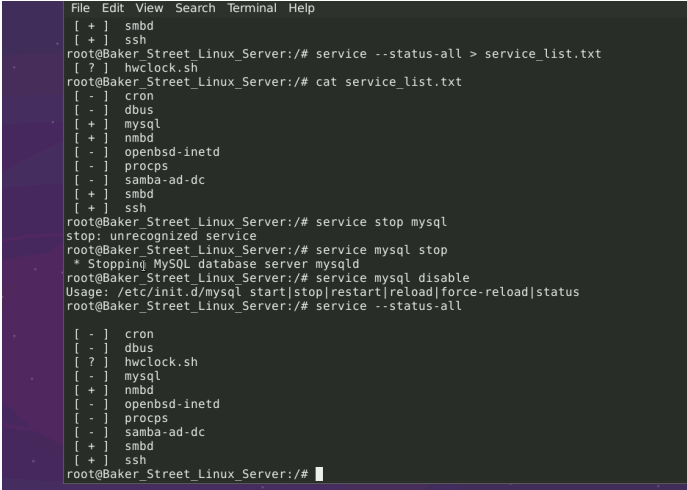
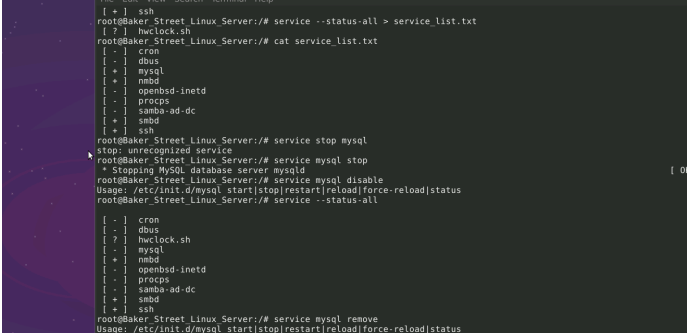
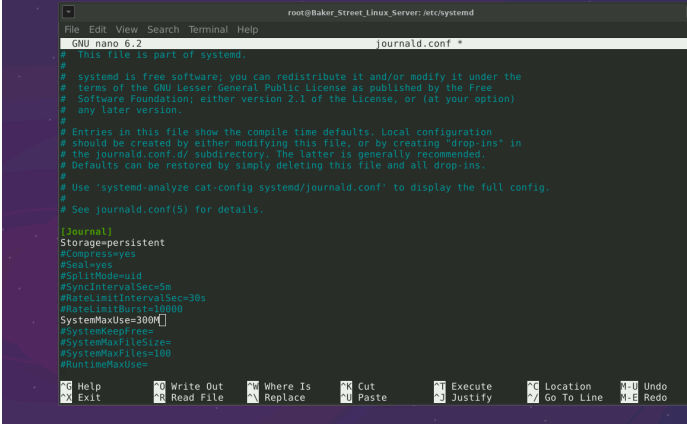
Dependencies removed

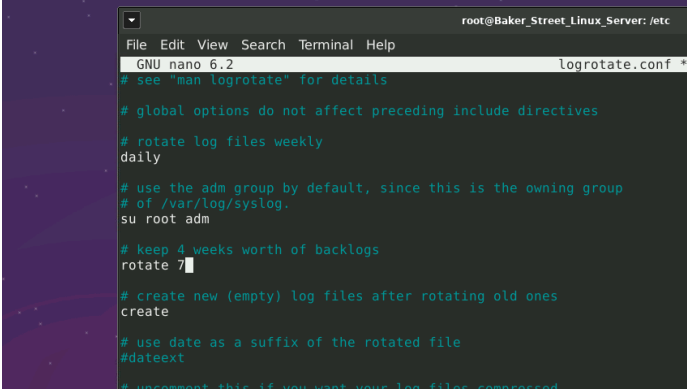
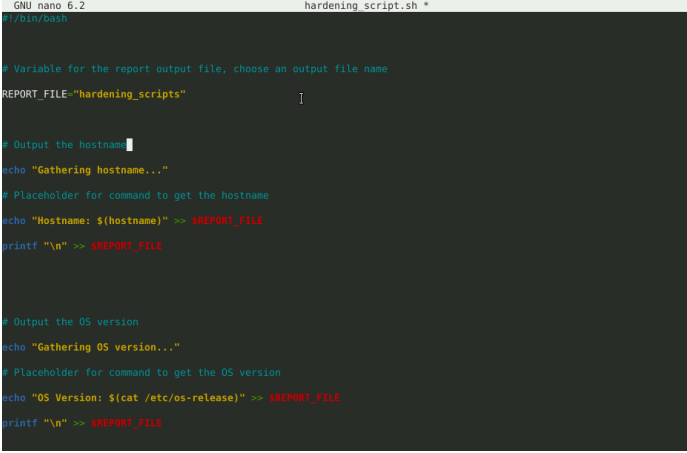
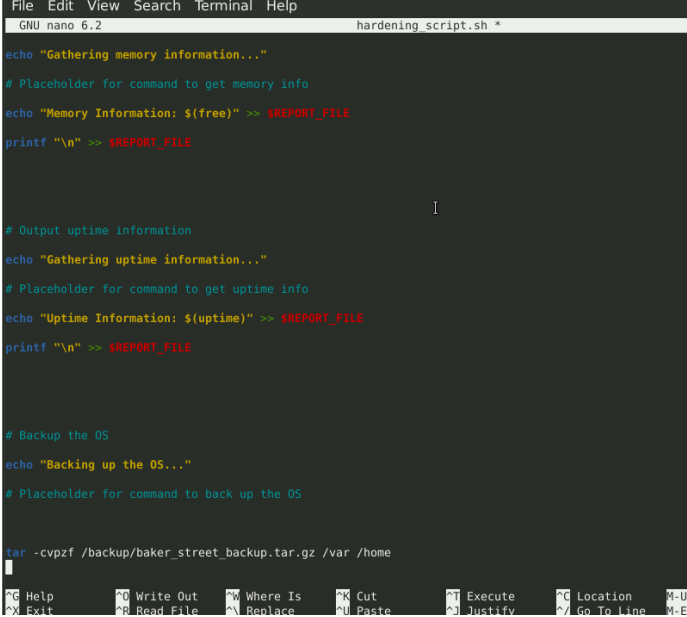
```
root@Baker_Street_Linux_Server:~# apt install ufw
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  iptables libip6tc2 libnetfilter_conntrack3 libnfnetlink0 libnftnl11
Suggested packages:
  firewallld kmod nftables
The following NEW packages will be installed:
  iptables libip6tc2 libnetfilter_conntrack3 libnfnetlink0 libnftnl11 ufw
0 upgraded, 6 newly installed, 0 to remove and 0 not upgraded.
Need to get 764 kB of archives.
After this operation, 4266 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libip6tc2 amd64 1.8.7-1ubuntu5.2 [20.3 kB]
Get:2 http://archive.ubuntu.com/ubuntu jammy/main amd64 libnfnetlink0 amd64 1.0.1-3build3 [14.6 kB]
Get:3 http://archive.ubuntu.com/ubuntu jammy/main amd64 libnetfilter_conntrack3 amd64 1.0.9-1 [45.3 kB]
Get:4 http://archive.ubuntu.com/ubuntu jammy/main amd64 libnftnl11 amd64 1.2.1-1build1 [65.5 kB]
Get:5 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 iptables amd64 1.8.7-1ubuntu5.2 [455 kB]
Get:6 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 ufw all 0.36.1-4ubuntu0.1 [162 kB]
Fetched 764 kB in 10s (76.0 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
```

Installed ufw

```
oot@Baker_Street_Linux_Server:~# apt install lynis -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  menu
Suggested packages:
  dnsutils apt-listbugs debsecan debsums tripwire samhain aide fail2ban menu-l10n gksu | kde-runtime
  | ktsuss
The following NEW packages will be installed:
  lynis menu
0 upgraded, 2 newly installed, 0 to remove and 0 not upgraded.
Need to get 581 kB of archives.
After this operation, 3164 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu jammy/universe amd64 lynis all 3.0.7-1 [227 kB]
Get:2 http://archive.ubuntu.com/ubuntu jammy/universe amd64 menu amd64 2.1.47ubuntu4 [354 kB]
Fetched 581 kB in 11s (53.4 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package lynis.
(Reading database ... 16659 files and directories currently installed.)
Preparing to unpack .../archives/lynis_3.0.7-1_all.deb ...
Unpacking lynis (3.0.7-1) ...
Selecting previously unselected package menu.
Preparing to unpack .../menu_2.1.47ubuntu4_amd64.deb ...
Unpacking menu (2.1.47ubuntu4) ...
Setting up lynis (3.0.7-1) ...
Created symlink /etc/systemd/system/timers.target.wants/lynis.timer → /lib/systemd/system/lynis.timer.
Setting up menu (2.1.47ubuntu4) ...
Processing triggers for menu (2.1.47ubuntu4) ...
root@Baker_Street_Linux_Server:~#
```


		Installed lynis with <pre> Created symlink /etc/systemd/system/timers.target.wants/lynis.timer → /lib/systemd/system/lynis.timer. Setting up menu (2:1.47ubuntu4) ... Processing triggers for menu (2:1.47ubuntu4) ... root@Baker_Street_Linux_Server:~# apt install tripwire -y Reading package lists... Done Building dependency tree... Done Reading state information... Done The following additional packages will be installed: cpio postfix ssl-cert Suggested packages: libarchive1v1 procmail postfix-mysql postfix-pgsql postfix-ldap postfix-pcre postfix-lmbd postfix-sqlite sasl2-bin dovecot-common resolvconf postfix-cdb mail-reader postfix-mta-sts-resolver postfix-doc The following NEW packages will be installed: cpio postfix ssl-cert tripwire 0 upgraded, 4 newly installed, 0 to remove and 0 not upgraded. Need to get 1109 kB of archives. </pre> <pre> setting relayhost: setting mynetworks: 127.0.0.0/8 [::ffff:127.0.0.0]/104 [::1]/128 setting mailbox_size_limit: 0 setting recipient_delimiter: + setting inet_interfaces: all setting inet_protocols: all /etc/aliases does not exist, creating it. WARNING: /etc/aliases exists, but does not have a root alias. Postfix (main.cf) is now set up with a default configuration. If you need to make changes, edit /etc/postfix/main.cf (and others) as needed. To view Postfix configuration values, see postconf(1). After modifying main.cf, be sure to run 'systemctl reload postfix'. Running newaliases invoke-rc.d: could not determine current runlevel invoke-rc.d: policy-rc.d denied execution of restart. invoke-rc.d: could not determine current runlevel invoke-rc.d: policy-rc.d denied execution of start. Created symlink /etc/systemd/system/multi-user.target.wants/postfix.service → /lib/sy service. Setting up tripwire (2.4.3.7-4) ... Generating site key (this may take several minutes)... Generating local key (this may take several minutes)... Processing triggers for ufw (0.36.1-4ubuntu0.1) ... Processing triggers for rsyslog (8.2112.0-2ubuntu2.2) ... invoke-rc.d: unknown initscript, /etc/init.d/rsyslog not found. invoke-rc.d: could not determine current runlevel root@Baker_Street_Linux_Server:~# </pre> Installed trip wire
☒	Disabling unnecessary services	Listed the services. They were then output into the service_list.txt. Mysql: Running Samba: Stopped <pre> cybersecurity/container project1 v4:latest ffad18376832757a3c59f6b9b03cc10466b4af905e549e19ae24391c90b073a root@Baker_Street_Linux_Server:~# cat /etc/passwd grep -v nologin sed 's/:\$/:\$@/' xargs sudo docker exec -it project1 v4 /bin/bash root@Baker_Street_Linux_Server:~# systemctl -t service --all System has not been booted with systemd as init system (PID 1). Can't operate. Failed to connect to bus: Host is down root@Baker_Street_Linux_Server:~# service --all --all: unrecognized service root@Baker_Street_Linux_Server:~# service --status-all [-] cron [-] dbus [7] hwclock.sh [+] mysql [+] nmbd [-] openbsd-inetd [-] procps [-] samba-ad-dc [+] smbd [+] ssh root@Baker_Street_Linux_Server:~# service --status-all > service_list.txt root@Baker_Street_Linux_Server:~# cat service_list.txt [-] cron [-] dbus [+] mysql [+] nmbd [-] openbsd-inetd [-] procps [-] samba-ad-dc [+] smbd [+] ssh root@Baker_Street_Linux_Server:~# </pre> Stopped mysql

		 <pre> File Edit View Search Terminal Help [+] smbd [+] ssh root@Baker_Street_Linux_Server:/# service --status-all > service_list.txt root@Baker_Street_Linux_Server:/# cat service_list.txt [-] cron [-] dbus [+] mysql [+] nmbd [-] openbsd-inetd [-] procps [-] samba-ad-dc [+] smbd [+] ssh root@Baker_Street_Linux_Server:/# service stop mysql stop: unrecognized service root@Baker_Street_Linux_Server:/# service mysql stop * Stopping MySQL database server mysqld root@Baker_Street_Linux_Server:/# service mysql disable Usage: /etc/init.d/mysql start stop restart reload force-reload status root@Baker_Street_Linux_Server:/# service --status-all [-] cron [-] dbus [?] hwclock.sh [-] mysql [+] nmbd [-] openbsd-inetd [-] procps [-] samba-ad-dc [+] smbd [+] ssh root@Baker_Street_Linux_Server:/# </pre>  <pre> [+] ssh root@Baker_Street_Linux_Server:/# service --status-all > service_list.txt [?] hwclock.sh root@Baker_Street_Linux_Server:/# cat service_list.txt [-] cron [-] dbus [+] mysql [+] nmbd [-] openbsd-inetd [-] procps [-] samba-ad-dc [+] smbd [+] ssh root@Baker_Street_Linux_Server:/# service stop mysql stop: unrecognized service root@Baker_Street_Linux_Server:/# service mysql stop * Stopping MySQL database server mysqld root@Baker_Street_Linux_Server:/# service mysql disable Usage: /etc/init.d/mysql start stop restart reload force-reload status root@Baker_Street_Linux_Server:/# service --status-all [-] cron [-] dbus [?] hwclock.sh [+] mysql [+] nmbd [-] openbsd-inetd [-] procps [-] samba-ad-dc [+] smbd [+] ssh root@Baker_Street_Linux_Server:/# service mysql remove Usage: /etc/init.d/mysql start stop restart reload force-reload status </pre>
☒	Enabling and configuring logging	 <pre> root@Baker_Street_Linux_Server:/etc/systemd File Edit View Search Terminal Help GNU nano 6.2 journald.conf # This file is part of systemd. # # systemd is free software; you can redistribute it and/or modify it under the # terms of the GNU Lesser General Public License as published by the Free # Software Foundation; either version 2.1 of the License, or (at your option) # any later version. # # Entries in this file show the compile time defaults. Local configuration # should be created by either modifying this file, or by creating "drop-ins" in # the journald.conf.d/ subdirectory. The latter is generally recommended. # Defaults can be restored by simply deleting this file and all drop-ins. # # Use 'systemd-analyze cat-config systemd/journald.conf' to display the full config. # # See journald.conf(5) for details. [journal] Storage=persistent #Compress=yes #Seal=yes #SslMode=auto #SyncIntervalSec=5m #RateLimitIntervalSec=30s #RateLimitBurst=10000 SystemMaxUse=300M #SystemKeepFree= #SystemMaxFiles= #SystemMaxFiles=100 #RuntimeMaxUse= </pre> Changed storage to persistent, set systemMaxUse to 300M.

		A screenshot of the GNU nano 6.2 text editor editing the file /etc/logrotate.conf. The terminal title bar shows 'root@Baker_Street_Linux_Server: /etc'. The editor menu includes File, Edit, View, Search, Terminal, and Help. The file content shows configuration for log rotation: '# global options do not affect preceding include directives', '# rotate log files weekly', 'daily', '# use the adm group by default, since this is the owning group of /var/log/syslog.', 'su root adm', '# keep 4 weeks worth of backlogs', 'rotate 7', '# create new (empty) log files after rotating old ones', 'create', '# use date as a suffix of the rotated file', '#dateext', and a commented line '# uncomment this if you want your log files compressed'. Changed log rotation to daily. Rotate out the logs after 7 days.
☒	Scripts created	A screenshot of the GNU nano 6.2 text editor editing the file hardening_script.sh. The terminal title bar shows 'GNU nano 6.2' and 'hardening_script.sh *'. The editor menu includes File, Edit, View, Search, Terminal, and Help. The file content shows the start of a script: '#!/bin/bash', '# Variable for the report output file, choose an output file name', 'REPORT_FILE="hardening_scripts"', '# Output the hostname', 'echo "Gathering hostname..."', '# Placeholder for command to get the hostname', 'echo "Hostname: \$(hostname)" >> \$REPORT_FILE', and 'printf "\n" >> \$REPORT_FILE'. Removed world permissions starting at /home. A screenshot of the GNU nano 6.2 text editor editing the file hardening_script.sh. The terminal title bar shows 'GNU nano 6.2' and 'hardening_script.sh *'. The editor menu includes File, Edit, View, Search, Terminal, and Help. The file content shows the continuation of the script: 'echo "Gathering memory information..."', '# Placeholder for command to get memory info', 'echo "Memory Information: \$(free)" >> \$REPORT_FILE', 'printf "\n" >> \$REPORT_FILE', '# Output uptime information', 'echo "Gathering uptime information..."', '# Placeholder for command to get uptime info', 'echo "Uptime Information: \$(uptime)" >> \$REPORT_FILE', 'printf "\n" >> \$REPORT_FILE', '# Backup the OS', 'echo "Backing up the OS..."', '# Placeholder for command to back up the OS', and 'tar -cvpzf /backup/baker_street_backup.tar.gz /var /home'. The bottom status bar shows various editor shortcuts like 'H Help', 'W Write Out', 'W Where Is', 'C Cut', 'E Execute', 'G Location', 'X Exit', 'R Read File', 'N Replace', 'P Paste', 'J Justify', 'G Go To Line', 'M-U', and 'M-E'. Removed world permissions starting at /home.

```

printf "\n" >> $REPORT_FILE

# Script to check for files with world permissions and update them
echo "Checking for files with world permissions..."

find /home -perm -o-w -exec chmod o-w {} \;

# Placeholder for command to find and update files with world permissions
echo "World permissions have been removed from any files found." >> $REPORT_FILE
printf "\n" >> $REPORT_FILE

# Find specific files and update their permissions

```

Placed command to update permissions for engineering, research and finance.

```

echo "Updating permissions for Engineering scripts."

# Placeholder for command to update permission
find -iname '*engineering*' -exec chown :engineering {} + -exec chmod 770 {} \;

echo "Permissions updated for Engineering scripts." >> $REPORT_FILE
printf "\n" >> $REPORT_FILE

```

```

find -iname '*research*' -exec chown :research {} + -exec chmod 770 {} \;

echo "Permissions updated for Research scripts" >> $REPORT_FILE
printf "\n" >> $REPORT_FILE

# Finance scripts - Only members of the finance group
echo "Updating permissions for Finance scripts"
# Placeholder for command to update permissions

find -iname '*finance*' -exec chown :finance {} + -exec chmod 770 {} \;

echo "Permissions updated for Finance scripts." >> $REPORT_FILE

```

```

# Variable for the report output file, choose a NEW output file name
REPORT_FILE="hardening_script2.txt"

# Output the sshd configuration file
echo "Gathering details from sshd configuration file"
# Placeholder for command to get the sshd configuration file

echo "sshd configuration file:$(cat /etc/ssh/sshd_config)" >> $REPORT_FILE
printf "\n" >> $REPORT_FILE

# Update packages and services
Echo ^^|Updating packages and services ^^|

```

Placed package commands to update/upgrade

```

GNU nano 6.2 hardening_script2.sh
sudo apt update >> $REPORT_FILE

# Placeholder for command to upgrade packages

sudo apt upgrade >> $REPORT_FILE

echo "Packages have been updated and upgraded" >> $REPORT_FILE
printf "\n" >> $REPORT_FILE

# Placeholder for command to list all installed packages

```

Added the package list command and output file

```

echo "Installed Packages:$(apt list --installed)" >> $REPORT_FILE
printf "\n" >> $REPORT_FILE

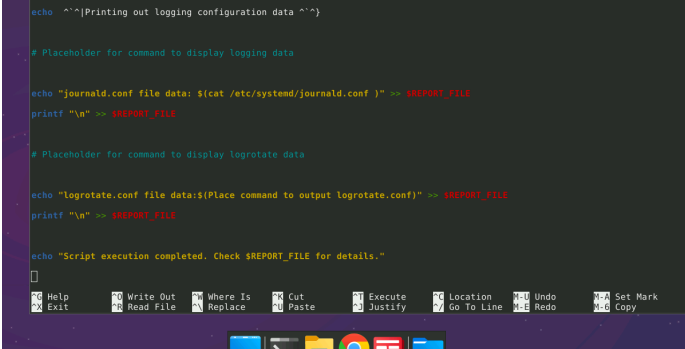
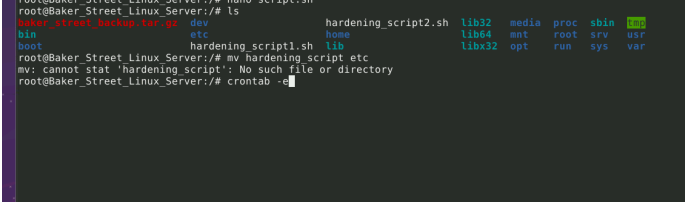
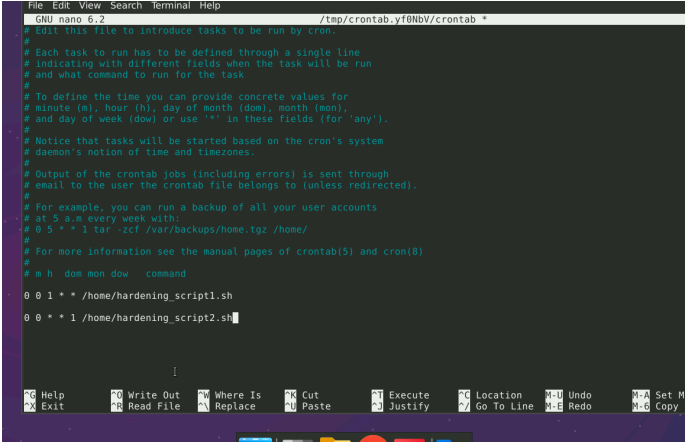
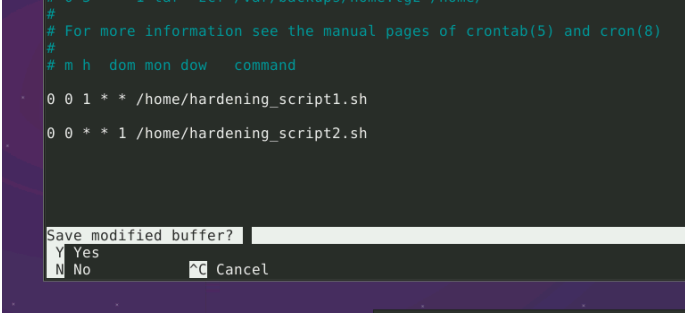
echo ^^|Printing out logging configuration data ^^|

# Placeholder for command to display logging data

echo "journald.conf file data: $(cat /etc/systemd/journald.conf )" >> $REPORT_FILE
printf "\n" >> $REPORT_FILE

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^L Location
^X Exit      ^R Read File  ^_ Replace    ^U Paste      ^J Justify    ^_ Go To Line

```

		 <pre> echo ^[[m(Printing out logging configuration data ^[[m) # Placeholder for command to display logging data echo "journal.conf file data: \$(cat /etc/systemd/journald.conf)" >> \$REPORT_FILE printf "\n" >> \$REPORT_FILE # Placeholder for command to display logrotate data echo "logrotate.conf file data:\$(Place command to output logrotate.conf)" >> \$REPORT_FILE printf "\n" >> \$REPORT_FILE echo "Script execution completed. Check \$REPORT_FILE for details." </pre>
☒	Scripts scheduled with cron	Scheduled to run the tasks repeatedly at specific times using crontab -e for both scripts.  <pre> root@Baker_Street_Linux_Server:/# ls root@Baker_Street_Linux_Server:/# mv hardening_script1.sh /home/hardening_script1.sh mv: cannot stat 'hardening_script1.sh': No such file or directory root@Baker_Street_Linux_Server:/# crontab -e </pre>  <pre> # Edit this file to introduce tasks to be run by cron. # # Each task to run has to be defined through a single line # indicating with different fields when the task will be run # and what command to run for the task # # To define the time you can provide concrete values for # minute (m), hour (h), day of month (dom), month (mon), # and day of week (dow) or use '*' in these fields (for 'any'). # # Notice that tasks will be started based on the cron's system # daemon's notion of time and timezones. # # Output of the crontab jobs (including errors) is sent through # email to the user the crontab file belongs to (unless redirected). # # For example, you can run a backup of all your user accounts # at 5 a.m. every week with: # 0 5 * * 1 tar -zcf /var/backups/home.tar.gz /home/ # # For more information see the manual pages of crontab(5) and cron(8) # # m h dom mon dow command 0 0 1 * * /home/hardening_script1.sh 0 0 * * 1 /home/hardening_script2.sh </pre> Script 1 Once a month on the 1st of the month Script 2 Once a week every Monday  <pre> # # For more information see the manual pages of crontab(5) and cron(8) # # m h dom mon dow command 0 0 1 * * /home/hardening_script1.sh 0 0 * * 1 /home/hardening_script2.sh </pre> Save modified buffer? [Y] Yes [N] No [^C] Cancel

		<pre># # For example, you can run a backup of all your user accounts # at 5 a.m every week with: # 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/ # # For more information see the manual pages of crontab(5) and cron(8) # # m h dom mon dow command 0 0 1 * * /home/hardening_script1.sh 0 0 * * 1 /home/hardening_script2.sh root@Baker_Street_Linux_Server:/#</pre>
--	--	--